# Supplementary Information: Genome-scale context, stoichiometric refinement, and neuro-symbolic reasoning for proteome function annotation

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## S1 Benchmark suite overview, Stage 0 calibration, and Suite 1 evaluations

Supplementary Table S1 summarizes the eight benchmark suites evaluated in the main manuscript, including their cohorts, Stage 0 evidence channel configurations, active pipeline stages, primary comparators, and canonical result tables in the repository. Supplementary Table S2 reports the ancestor-propagated evaluation across the 13-genome and 21-genome reference panels comparing GSPA against production annotation pipelines, individual Stage 0 channels, and genomic language models.

**Table S1. Overview of the eight GSPA evaluation suites, cohorts, Stage 0 channel configurations, active pipeline stages, and canonical result tables.**

| Suite | Evaluation Scenario & Cohort | Proteomes / Proteins | Stage 0 Channels | Active Stages | Primary Comparators | Canonical Result Table(s) |
|---|---|---|---|---|---|---|
| Suite 1A | CAFA3→CAFA5 Temporal GO Holdout | 16,125 evals (3,440 at < 30% ID) | 5-Ch (DIAMOND, Pfam, InterPro, ESM-2, ProtT5) | 0 + 1 + 2A | DIAMOND, ProteInfer, DeepGOCNN, DeepGOPlus, DeepGOZero | scenario1_cafa_temporal_holdout_real.tsv |
| Suite 1B | Prospective CAFA6 Sequence Holdout | 769 prospective targets | 7-Ch (ESM-2, ProstT5 3Di, DIAMOND, InterPro, DeepGO-SE) | 0 + 2A + 3 | ESM-2+3Di MLP, Qwen3-14B Agent | suite1_cafa6_prospective_summary.tsv |
| Suite 1C | Post-Cutoff Swiss-Prot EC (Holdout-125) | 125 enzymes (74 species, 293 pairs) | 1-Ch or 3-Ch (CLEAN, DeepProZyme, EnzBERT) | 0 + 1 + 2B | CLEAN, DeepProZyme, EnzBERT, METEOR v8 | scenario1_swissprot_ec_holdout125_real.tsv |
| Suite 1D | 95-Genome Bacterial EC Homology Masking | 95 genomes (69,666 proteins; 4 ID tiers) | 1-Ch or 3-Ch (CLEAN, DeepProZyme, EnzBERT) | 0 + 1 + 2B | Standalone Predictors, METEOR v8 | scenario1_95genomes_ec_homology_real.tsv |
| Suite 2A | 9-Genome Stage 1/2A Ablation & Perturbation | 9 genomes (81 runs; 900 injected claims) | 3-Ch (DIAMOND, HMMER/Pfam, InterProScan) | 0 + 1 + 2A | Stage 0 Noisy-OR, Individual Priors $B_1$–$B_6$ | pillar1_2_ablation_summary.tsv |
| Suite 2B | 13/21-Genome SOTA & 88-Proteome Multi-Kingdom | 13, 21, and 88 proteomes (4 domains) | 3-Ch or 7-Ch (Full Neural + Structural) | 0 + 1 + 2A | NCBI PGAP, InterProScan, deepFRI, gLM, gLM2 | pillar1_suite2_panel13_21_sota.tsv; multikingdom_summary_by_domain.tsv |
| Suite 2C | 1,000-Genome Bacterial Population Audit | 1,000 RefSeq bacterial genomes | 3-Ch (DIAMOND, HMMER/Pfam, InterProScan) | 0 + 1 + 2A | DeepGO-SE, SPROF-GO, DeepGraphGO, DeepFRI, TALE, InterProScan | pillar1_suite2_1000bacteria_sota.tsv |
| Suite 2D | 6-Eukaryote Anatomical Bridge & Relaxation | 6 species (6,155 endog.; 7,655 stress) | 7-Ch (Full Neural + Structural) | 0 + 1 + 2A | Direct GO Taxon Rules, Stage 2A Term Deletion | pillar3_eukaryote_anatomy.tsv; pillar3_relax_vs_remove.tsv |
| Suites 3–5 | Role-Stratified LRO & Heteromeric Complex LSO | 256/414 (*E. coli*), 144 (*B. subtilis*), 175 LSO | Track A (No PLM) or Track B (+ ESM-2 Centroids) | 0 + 1 + 2B + 3 | Sequence-Only/METEOR, RLGC Alone, GBDT | pillar2_lro15_summary.tsv; pillar3_complex_lso.tsv |
| Suite 6 | RB-TnSeq Fitness Enzymes (Price-149) & 108-Audit | 148 proteins (22 hosts); 108 GEMs; 6 BiGG | 1-Ch or 5-Ch (CLEAN, MAPred, DPZ, EnzBERT, GraphEC, TopEC) | 0 + 1 + 2B | 6 Standalone Tools, METEOR v8, DCGM, ECtocer | pillar2_price149_all6_real.tsv; pillar2_invariant_audit_summary.tsv |
| Suites 7–8 | BacDive Carbon Growth & Diagnostic Enzymes | 529 (2,332), 3,017 (17,496), 6,313 (50,075) | 3-Model Pool (gapseq, ModelSEED, CarveMe) | 0 + 1 + 2B | CarveMe, ModelSEED, gapseq, METEOR, Traitar, DiASPora | pillar3_bacdive_track1_summary.tsv; pillar3_bacdive_track2_summary.tsv |
| Stage 3 | De-Identified 120-Anomaly & 50-LRO Benchmark | 120 anomalies (7 taxa); 50 LRO gaps (3 seeds) | Post-Stage-2 Candidate Set | 1 + 2 + 3 | Threshold ≥ 0.35, 5-Fold CV LR/RF/GBDT/MLP, No-Guardrail | pillar4_agent_evaluation.tsv; pillar4_multiseed_summary.tsv |

**Table S2. Ancestor-propagated evaluation across the 13-genome prokaryotic panel (11 prokaryotes), 21-genome multi-kingdom panel, and 9-genome genomic language model subset.**

| Panel Cohort | Method / Channel | Genomes ($n$) | Micro $F_{\max}$ | CAFA $F_{\max}$ | $\mathbf{WF}_{\max}$ | $\mathbf{S}_{\min}$ | EC $F_{\max}$ | Runtime / Peak RAM |
|---|---|---|---|---|---|---|---|---|
| 13-Genome Panel (Prok. 11) | metagenomic-deepFRI (2026) | 11 | 0.6010 | 0.5940 | 0.1565 | 95.20 | n/a | 38.0 s / 4,800 MB |
| | InterProScan 5.72 | 11 | 0.7410 | 0.6780 | 0.4015 | 58.40 | n/a | 680.0 s / 8,200 MB |
| | NCBI PGAP (2021) | 11 | 0.7820 | 0.7680 | 0.4438 | 54.20 | n/a | 1,800.0 s / 16,000 MB |
| | **GSPA Stage 1+2 (3-Channel Full)** | 11 | 0.8421 | 0.8605 | 0.7215 | 21.40 | 0.8830 | 5.4 s / 4,812 MB |
| 21-Genome Panel (All 21) | FoldSeek + ProstT5 3Di (2024) | 21 | 0.6840 | 0.6310 | n/a | 83.24 | n/a | 10,800.0 s / 14,000 MB |
| | InterProScan 5.72 | 21 | 0.7180 | 0.6490 | n/a | 95.93 | 0.3120 | 720.0 s / 8,200 MB |
| | ESM2-DeepGOPlus (2024) | 21 | 0.7420 | 0.6950 | n/a | 105.29 | n/a | 420.0 s / 6,400 MB |
| | ProteInfer (2023) | 21 | 0.7950 | 0.7540 | n/a | 38.12 | 0.3880 | 120.0 s / 5,800 MB |
| | DIAMOND-BLASTp | 21 | 0.8400 | 0.7890 | n/a | 93.81 | 0.8550 | 17.0 s / 2,100 MB |
| | ProtT5 $k$-NN (2021) | 21 | 0.8530 | 0.8010 | n/a | 39.40 | n/a | 28.0 s / 6,200 MB |
| | ESM-2 $k$-NN (2023) | 21 | 0.8580 | 0.8080 | n/a | 37.85 | n/a | 24.0 s / 5,800 MB |
| | **GSPA Stage 1+2 (3-Channel Core)** | 21 | 0.8690 | 0.8220 | n/a | 36.14 | 0.8830 | 6.1 s / 4,812 MB |
| | **GSPA Stage 1+2 (7-Channel All-Pooled)** | 21 | 0.8705 | 0.8407 | n/a | 35.20 | 0.8860 | 6.8 s / 4,812 MB |
| 9-Genome Subset | gLM / gLM2 Genomic LM (2024) | 9 | 0.7280 | 0.7610 | n/a | 37.40 | n/a | 95.0 s / 7,200 MB |

### S1.1 Stage 0 Platt calibration parameters and correlation groups

In Stage 0 (`ScoreCalibrator.groovy` and `LikelihoodModel.groovy`), raw predictor scores $s(c) \in [0, 1]$ are mapped via Platt scaling $\tilde{q}(c) = \text{clip}(\sigma_{\log}(a_e s(c) + b_e), 10^{-4}, 0.9999)$ and multiplied by channel reliability $r_e \in (0, 1]$. Default calibration slopes and intercepts $(a_e, b_e)$ are $(7.0, -3.5)$ for sequence alignment (DIAMOND/BLAST), $(6.5, -2.9)$ for profile HMMs (HMMER/Pfam), $(6.0, -2.4)$ for InterProScan and KofamScan, $(6.8, -3.2)$ for ESM-2 and ProtT5 $k$-NN transfer, $(6.2, -2.8)$ for ProteInfer, DeepGOWeb, and DeepGO-SE, and $(6.4, -3.0)$ for FoldSeek and ProstT5 3Di structural transfer. Default channel reliability weights $r_e$ (`EvidenceType.groovy`) are 0.85 for AMR domain predictors, 0.80 for sequence homology and orthology, 0.75 for profile HMMs and sequence domains, 0.70 for protein-sequence ML, PLM $k$-NN, and structural similarity, 0.65 for structure-based deep learning, 0.60 for metabolic pathway context and inferred context, 0.55 for genomic language models, and 0.45 for raw genomic neighborhood context. Claims are partitioned into 11 correlation groups in `LikelihoodModel.groovy`: homology, protein-sequence ML, structure, genomic context, genomic language models, inferred context, sequence motifs, localization, region, viral, and specialized domain predictors.

## S2 Hyperparameter specification and Stage 1–2A implementation crosswalk

All 18 configurable pipeline hyperparameters across Stage 1 (13 configuration keys exposed by `IntegrateCommand.groovy` and consumed by `IterativeRefiner.groovy` and the Stage 1 prior modules, alongside the fixed candidate partner gate 0.08 in `CoherencePrior.groovy`, the fixed co-operonic complex partner bonus 0.95 in `GenomicContextPrior.groovy`, the Stage 2A ancestor relaxation factor 1.0 in `ConsistencyEnforcer.groovy`, and the Stage 2A obligate-complex singleton prune threshold 0.45 in `IntegrateCommand.groovy`) and Stage 2B (5 parameters in `stoichiometric_locus_refiner.py`, alongside the fixed elemental-imbalance penalty $w_{\text{imb}} = 100.0$) were calibrated on the *Escherichia coli* K-12 MG1655 development proteome and held frozen across all out-of-sample prokaryotic, archaeal, viral, and eukaryotic benchmarks. Supplementary Table S3 reports these 18 configurable hyperparameters, their configuration keys, source modules, roles in inference, and default values (where penalty parameters are stored as positive magnitudes and applied with negative sign in log-odds space).

### S2.1 Temporal GO and EC holdouts and 95-genome homology masking

Supplementary Table S4 reports the CAFA temporal holdout benchmark across 16,125 protein–aspect evaluations (3,440 at $< 30\%$ sequence identity). Supplementary Table S5 reports the 16-row post-cutoff Swiss-Prot enzyme benchmark (Holdout-125, $n = 125$ proteins across 293 ground-truth protein–EC pairs), separating standalone per-protein baselines, single-predictor refinement, and three-method ensemble pooling. Supplementary Table S6 reports the 12-row enzyme evaluation across the 95-genome bacterial panel under four donor-database homology-masking tiers (unfiltered, $\leq 70\%$, $\leq 50\%$, and $\leq 30\%$ sequence identity), comparing unfiltered METEOR v8 Top-1 changes against GSPA's single-predictor stoichiometric and pathway acceptance filter and GSPA's three-channel consensus acceptance filter, alongside exact discordant protein counts $(b_{01}, b_{10})$ and two-sided exact McNemar $p$-values.

**Table S3. Complete inventory of the 18 configurable GSPA v2.1.0 hyperparameters across Stages 1 and 2B.**

| Stage / Module | Symbol | Parameter Key | Source Module | Role in Inference | Default Value |
|---|---|---|---|---|---|
| Stage 1 ($B_1$) | $\alpha_{\text{ess}}$ | `alpha_ess` | `EssentialityPrior.groovy` | Unfulfilled essential GO function log-odds boost | 1.50 |
| Stage 1 ($B_2$) | $\alpha_{\text{coh}}$ | `alpha_coh` | `CoherencePrior.groovy` | Active pathway coherence log-odds scale | 1.00 |
| Stage 1 ($B_2$) | $w_{\text{part}}$ | `alpha_complex_boost` | `CoherencePrior.groovy` | Obligate heteromeric complex partner boost ($0.65\,w_{\text{part}} = 0.5525$ for $N_{\text{strong}} \geq 2$) | 0.85 |
| Stage 1 ($B_2$) | $w_{\text{sing}}$ | `alpha_complex_penalty` | `CoherencePrior.groovy` | Isolated obligate heteromeric singleton penalty magnitude (applied as $-0.75$) | 0.75 |
| Stage 1 ($B_3$) | $\alpha_{\text{cons}}$ | `alpha_cons` | `ConsistencyPrior.groovy` | SAT4J taxonomic conflict penalty magnitude (applied as $-0.68$) | 0.68 |
| Stage 1 ($B_4$) | $\alpha_{\text{anat}}$ | `alpha_anat_penalty` | `AnatomicalPrior.groovy` | Taxon-forbidden Uberon/PO/CL structure penalty magnitude (applied as $-0.68$) | 0.68 |
| Stage 1 ($B_4$) | $\alpha_{\text{boost}}$ | `alpha_anat_boost` | `AnatomicalPrior.groovy` | Active-structure developmental rescue log-odds boost | 0.65 |
| Stage 1 ($B_5$) | $\alpha_{\text{gap}}$ | `alpha_gap` | `GapFillingPrior.groovy` | Metabolic pathway hole-filling log-odds boost | 1.50 |
| Stage 1 ($B_6$) | $\alpha_{\text{ctx}}$ | `alpha_ctx` | `GenomicContextPrior.groovy` | Co-operonic pathway enrichment log-odds scale | 1.00 |
| Stage 1 ($B_6$) | $\alpha_{\text{gap_ctx}}$ | `alpha_gap_ctx` | `GenomicContextPrior.groovy` | Co-operonic metabolic gap multiplier on $\alpha_{\text{ctx}}\, c_O(W)$ | 1.50 |
| Stage 1 (Iter) | $\gamma_{\text{iter}}$ | `damping` | `IterativeRefiner.groovy` | Iterative log-odds fixed-point damping weight | 0.50 |
| Stage 1 (Iter) | $K_{\max}$ | `max_iter` | `IterativeRefiner.groovy` | Maximum Stage 1 message-passing iterations | 6 |
| Stage 1 (Iter) | $\epsilon$ | `epsilon` | `IterativeRefiner.groovy` | Stage 1 probability convergence tolerance | 0.005 |
| Stage 2B (SLR) | $\tau_{\text{core}}$ | `tau_core` | `stoichiometric_locus_refiner.py` | Core reaction probability threshold $S_{\text{rxn}}(j)$ | 0.50 |
| Stage 2B (SLR) | $\alpha_{\text{bio}}$ | `alpha_bio` | `stoichiometric_locus_refiner.py` | Minimum required fraction of $v_{\text{bio}}^{\max}$ in pFBA | 0.10 |
| Stage 2B (SLR) | $\eta_{\text{null}}$ | `tau_null` | `stoichiometric_locus_refiner.py` | Unassigned-gene hypothesis $h_{\varnothing}$ prior mass | 0.05 |
| Stage 2B (SLR) | $\gamma_{\text{FBA}}$ | `gamma_fba` | `stoichiometric_locus_refiner.py` | Backward locus responsibility log-odds scale | 2.20 |
| Stage 2B (SLR) | $\gamma_{\text{damp}}$ | `gamma_damp` | `stoichiometric_locus_refiner.py` | Geometric damping decay across outer passes | 0.60 |

**Table S4. CAFA temporal holdout evaluation across 16,125 protein–aspect evaluations (3,440 twilight-zone evaluations at $< 30\%$ sequence identity).**

| Method | MF All | MF $< 30\%$ | BP All | BP $< 30\%$ | CC All | CC $< 30\%$ | Macro $F_{\max}$ All | Macro $F_{\max} < 30\%$ | Runtime / Peak RAM |
|---|---|---|---|---|---|---|---|---|---|
| DIAMOND (BLAST) | 0.6224 | 0.0000 | 0.4437 | 0.0000 | 0.5805 | 0.0000 | 0.5489 | 0.0000 | 4.2 s / 1,150 MB |
| DeepGOCNN | 0.4280 | 0.4209 | 0.3431 | 0.3389 | 0.6409 | 0.6487 | 0.4707 | 0.4695 | 8.5 s / 2,400 MB |
| DeepGOPlus (CNN+BLAST) | 0.6337 | 0.4207 | 0.4613 | 0.3364 | 0.6722 | 0.6468 | 0.5891 | 0.4680 | 12.7 s / 2,650 MB |
| DeepGOZero | 0.6552 | 0.5006 | 0.4514 | 0.3614 | 0.6602 | 0.6327 | 0.5889 | 0.4982 | 9.4 s / 2,800 MB |
| InterPro MLP | 0.6561 | 0.5066 | 0.4592 | 0.3707 | 0.6667 | 0.6384 | 0.5940 | 0.5052 | 14.2 s / 3,100 MB |
| DeepGOZero+BLAST | 0.6676 | 0.4982 | 0.4816 | 0.3614 | 0.6668 | 0.6327 | 0.6053 | 0.4974 | 13.6 s / 2,950 MB |
| ProteInfer | 0.6966 | 0.6810 | 0.6146 | 0.5948 | 0.5301 | 0.5181 | 0.6138 | 0.5980 | 38.4 s / 6,400 MB |
| Stage 0 Noisy-OR (Ensemble) | 0.7159 | 0.6949 | 0.6387 | 0.6144 | 0.7241 | 0.7075 | 0.6929 | 0.6723 | 56.8 s / 6,400 MB |
| **GSPA (Iter 1)** | 0.7163 | 0.6971 | 0.6391 | 0.6152 | 0.7257 | 0.7202 | 0.6937 | 0.6775 | 59.2 s / 6,450 MB |
| GSPA (Iter 2) | 0.7164 | 0.6972 | 0.6365 | 0.6143 | 0.7259 | 0.7202 | 0.6929 | 0.6772 | 61.5 s / 6,450 MB |
| GSPA (Iter 3) | 0.7165 | 0.6972 | 0.6332 | 0.6104 | 0.7259 | 0.7202 | 0.6919 | 0.6759 | 63.8 s / 6,450 MB |

**Table S5. Complete post-cutoff Swiss-Prot enzyme evaluation (Holdout-125, $n = 125$ proteins, 293 ground-truth EC annotations).**

| Predictor / Refinement Method | Masking Tier | Top-1 Acc. (Hits) | Top-3 Acc. (Hits) | Top-5 Acc. (Hits) | MRR | Micro $F_{\max}$ |
|---|---|---|---|---|---|---|
| CLEAN (2023) | Unfiltered | 32.8% (41/125) | 40.8% (51/125) | 41.6% (52/125) | 0.3719 | 0.4360 |
| DeepProZyme (2023) | Unfiltered | 23.2% (29/125) | 31.2% (39/125) | 32.0% (40/125) | 0.2750 | 0.2715 |
| DeepProZyme + METEOR v8 (Single-Pred) | Unfiltered | 25.6% (32/125) | 31.2% (39/125) | 31.2% (39/125) | 0.2872 | 0.2804 |
| DeepProZyme + GSPA Single-Pred (SLR + Coh.) | Unfiltered | 23.2% (29/125) | 31.2% (39/125) | 32.8% (41/125) | 0.2779 | 0.2703 |
| EnzBERT (2023) | Unfiltered | 32.8% (41/125) | 37.6% (47/125) | 39.2% (49/125) | 0.3531 | 0.3596 |
| Noisy-OR (3-Method Ensemble) | Unfiltered | 33.6% (42/125) | 40.8% (51/125) | 43.2% (54/125) | 0.3807 | 0.4381 |
| GSPA (3-Method, Iter 1) | Unfiltered | 32.0% (40/125) | 42.4% (53/125) | 44.0% (55/125) | 0.3775 | 0.4035 |
| GSPA (3-Method, Iter 2) | Unfiltered | 32.0% (40/125) | 42.4% (53/125) | 44.0% (55/125) | 0.3775 | 0.3891 |
| GSPA (3-Method, Iter 3) | Unfiltered | 32.0% (40/125) | 43.2% (54/125) | 44.0% (55/125) | 0.3781 | 0.3830 |
| EnzBERT | $\leq 70\%$ identity | 30.4% (38/125) | 35.2% (44/125) | 35.2% (44/125) | 0.3305 | 0.3204 |
| EnzBERT | $\leq 50\%$ identity | 28.0% (35/125) | 33.6% (42/125) | 33.6% (42/125) | 0.3040 | 0.3163 |
| EnzBERT | $\leq 30\%$ identity | 17.6% (22/125) | 24.8% (31/125) | 24.8% (31/125) | 0.2139 | 0.2212 |
| CLEAN | $\leq 30\%$ identity | 24.0% (30/125) | 30.4% (38/125) | 32.8% (41/125) | 0.2741 | 0.2299 |
| DeepProZyme | $\leq 70\%$ identity | 23.2% (29/125) | 31.2% (39/125) | 32.0% (40/125) | 0.2713 | 0.2715 |
| DeepProZyme | $\leq 50\%$ identity | 23.2% (29/125) | 31.2% (39/125) | 32.8% (41/125) | 0.2720 | 0.2684 |
| DeepProZyme | $\leq 30\%$ identity | 22.4% (28/125) | 31.2% (39/125) | 32.8% (41/125) | 0.2674 | 0.2650 |

**Table S6. Single-predictor and 3-channel consensus acceptance filtering of candidate METEOR v8 EC changes across 95 bacterial genomes under four donor-database homology-masking tiers.** $p$-values report two-sided exact binomial McNemar tests on discordant Top-1 predictions ($b_{01}/b_{10}$).

| Predictor | Tier | Genomes / Proteins | Base Top-1 ($F_{\max}$) | METEOR v8 Top-1 ($\Delta$) | METEOR $b_{01}/b_{10}$ ($p$) | GSPA Single-Pred Top-1 ($\Delta$) | Single-Pred $b_{01}/b_{10}$ ($p$) | GSPA Multi-Pred Top-1 ($\Delta$) | Multi-Pred $b_{01}/b_{10}$ ($F_{\max}$) |
|---|---|---|---|---|---|---|---|---|---|
| CLEAN | Unfiltered | 73/52,814 | 0.8214 (0.8796) | 0.8234 (+108) | 184/76 ($< 10^{-6}$) | 0.8228 (+77) | 99/22 ($< 10^{-6}$) | 0.8243 (+155) | 162/7 (0.8816) |
| CLEAN | $\leq 70\%$ | 95/69,663 | 0.8267 (0.8836) | 0.8289 (+149) | 191/42 ($< 10^{-6}$) | 0.8283 (+108) | 131/23 ($< 10^{-6}$) | 0.8289 (+149) | 154/5 (0.8851) |
| CLEAN | $\leq 50\%$ | 95/69,663 | 0.8271 (0.8844) | 0.8294 (+162) | 208/46 ($< 10^{-6}$) | 0.8291 (+139) | 156/17 ($< 10^{-6}$) | 0.8293 (+155) | 158/3 (0.8859) |
| CLEAN | $\leq 30\%$ | 94/68,681 | 0.8288 (0.8833) | 0.8306 (+122) | 156/34 ($< 10^{-6}$) | 0.8302 (+93) | 107/14 ($< 10^{-6}$) | 0.8306 (+126) | 132/6 (0.8845) |
| DeepProZyme | Unfiltered | 95/69,666 | 0.9022 (0.9331) | 0.9010 (−85) | 103/188 ($1.0 \times 10^{-6}$) | 0.9017 (−33) | 24/57 ($3.2 \times 10^{-4}$) | 0.9027 (+36) | 51/15 (0.9335) |
| DeepProZyme | $\leq 70\%$ | 95/69,666 | 0.8965 (0.9290) | 0.8961 (−27) | 151/178 (0.152) | 0.8963 (−15) | 44/59 (0.167) | 0.8975 (+67) | 78/11 (0.9296) |
| DeepProZyme | $\leq 50\%$ | 95/69,666 | 0.8966 (0.9288) | 0.8966 (+1) | 163/162 (1.000) | 0.8966 (−3) | 49/52 (0.842) | 0.8979 (+87) | 93/6 (0.9297) |
| DeepProZyme | $\leq 30\%$ | 95/69,666 | 0.8967 (0.9289) | 0.8970 (+19) | 170/151 (0.315) | 0.8966 (−4) | 51/55 (0.771) | 0.8976 (+67) | 76/9 (0.9296) |
| EnzBERT | Unfiltered | 95/69,666 | 0.8430 (0.8707) | 0.8416 (−92) | 100/192 ($< 10^{-6}$) | 0.8428 (−10) | 51/61 (0.395) | 0.8438 (+59) | 65/6 (0.8713) |
| EnzBERT | $\leq 70\%$ | 95/69,666 | 0.8394 (0.8675) | 0.8374 (−146) | 96/242 ($< 10^{-6}$) | 0.8391 (−26) | 41/67 (0.0157) | 0.8405 (+70) | 76/6 (0.8682) |
| EnzBERT | $\leq 50\%$ | 95/69,666 | 0.8330 (0.8602) | 0.8304 (−181) | 132/313 ($< 10^{-6}$) | 0.8324 (−38) | 31/69 ($1.8 \times 10^{-4}$) | 0.8343 (+89) | 93/4 (0.8611) |
| EnzBERT | $\leq 30\%$ | 95/69,666 | 0.8256 (0.8515) | 0.8232 (−167) | 108/275 ($< 10^{-6}$) | 0.8254 (−18) | 39/57 (0.082) | 0.8267 (+78) | 80/2 (0.8524) |

# S3 Cross-ontology bridge and multi-kingdom population evaluations

Supplementary Table S7 summarizes the five resource tables programmatically extracted from the official OBO Foundry releases of the Gene Ontology (GO), Uberon multi-species anatomy ontology, Plant Ontology (PO), Cell Ontology (CL), and NCBI Taxonomy and loaded at runtime by the taxonomic and anatomical constraint modules. Supplementary Table S8 reports the 1,000-genome RefSeq bacterial population audit, and Supplementary Table S9 reports the 88-proteome multi-kingdom evaluation across Archaea, Bacteria, Eukaryota, and Viruses.

**Table S7. Summary of programmatic cross-ontology constraint tables in GSPA v2.1.0.**

| Resource File | Ontologies Bridged | Data Rows | Distinct Classes | Constraint Semantics & Examples |
|---|---|---|---|---|
| `go-taxon-constraints.tsv` | GO ↔ NCBITaxon | 26,911 | 18,159 GO terms | 13,819 `only_in_taxon` + 13,092 `never_in_taxon` rules (e.g., `GO:0009579` thylakoid) |
| `go-anatomical-constraints.tsv` | GO ↔ Uberon / PO / CL | 2,715 | 2,674 GO; 1,294 Anat. | 30 object properties (`results_in_development_of`, `occurs_in`, etc.) |
| `uberon-po-taxon-constraints.tsv` | Uberon / PO / CL ↔ NCBITaxon | 8,331 | 1,247 Anat. classes | 4,767 `only_in_taxon` + 3,564 `never_in_taxon` rules (e.g., `UBERON:0000948` heart → Metazoa) |
| `uberon-po-hierarchy.tsv` | Uberon / PO / CL DAG | 7,722 | 2,855 Anat. classes | Transitive `is_a` and `part_of` edges across anatomical structures |
| `ncbi-taxon-hierarchy.tsv` | NCBI Taxonomy DAG | 1,818 | 1,305 Taxon clades | Transitive taxonomic `is_a` lineage closure for benchmark clades |

**Table S8. Population-scale structural audit across 1,000 RefSeq bacterial genomes comparing deep-learning GO predictors, InterProScan, Stage 0 pooling, and full GSPA.**

| Method | Genomes ($n$) | Macro $F_{\max}$ | Essential Core-22 (%) | Essential Full-29 (%) | Complex Coh. (%) | MetaCyc Comp. (%) | Taxon Viol. / Genome | Runtime / Peak RAM |
|---|---|---|---|---|---|---|---|---|
| DeepGO-SE / DeepGOMeta (2024) | 1,000 | 0.5047 | 95.34% | 82.17% | 79.83% | 11.59% | 14,198.8 | 14.2 s / 4,200 MB |
| SPROF-GO (2023) | 1,000 | 0.4930 | 99.91% | 83.70% | 79.18% | 0.18% | 7,322.7 | 28.5 s / 6,100 MB |
| DeepGraphGO (2021) | 1,000 | 0.4020 | 99.28% | 85.72% | 70.56% | 37.37% | 9,922.2 | 19.4 s / 5,400 MB |
| DeepFRI (2021) | 1,000 | 0.3843 | 100.00% | 81.25% | 98.87% | 9.43% | 106,815.2 | 24.0 s / 4,800 MB |
| TALE (2021) | 1,000 | 0.3883 | 87.24% | 75.38% | 65.73% | 19.63% | 9,794.7 | 16.8 s / 4,500 MB |
| InterProScan 5 | 1,000 | 0.5940 | 97.36% | 86.90% | 61.89% | 40.35% | 940.4 | 145.0 s / 8,200 MB |
| GSPA Stage 0 (Noisy-OR) | 1,000 | 0.6929 | 99.81% | 90.42% | 66.03% | 36.71% | 23,384.1 | 0.85 s / 3,400 MB |
| **GSPA Stage 1+2 (Full)** | 1,000 | 0.6937 | 99.99% | 95.29% | 100.00% | 66.75% | 0.0 | 5.4 s / 4,812 MB |

**Table S9. Multi-kingdom proteome-wide evaluation across 88 proteomes (121,631 proteins) spanning Archaea, Bacteria, Eukaryota, and Viruses under homology masking.**

| Domain | Configuration | Genomes / Proteins | Mean $F_{\max}$ (All) | Mean $F_{\max}$ (< 30%) | MF / BP / CC $F_{\max}$ | Coverage | $\Delta F_{\max}$ [95% Bootstrap CI] | Runtime / Heap | Stage 3 Trigger (%) |
|---|---|---|---|---|---|---|---|---|---|
| Archaea | Stage 0 Noisy-OR | 15/8,639 | 0.9125 | 0.5419 | 0.9460/0.9341/0.9146 | 0.9524 | Baseline | 0.35 s / 4,746 MB | 1.97% |
| | **GSPA Stage 1+2A** | 15/8,639 | 0.9129 | 0.5422 | 0.9454/**0.9345**/**0.9163** | 0.9524 | +0.0004 [−0.0001, +0.0010] | 2.29 s / 5,120 MB | 1.97% |
| Bacteria | Stage 0 Noisy-OR | 47/104,032 | 0.7987 | 0.7987 | 0.8326/0.8304/0.8322 | 0.8443 | Baseline | 2.16 s / 4,479 MB | 1.72% |
| | **GSPA Stage 1+2A** | 47/104,032 | 0.8009 | 0.8009 | 0.8319/**0.8316**/**0.8371** | 0.8443 | +0.0022 [+0.0014, +0.0031] | 12.66 s / 4,812 MB | 1.72% |
| Eukaryota | Stage 0 Noisy-OR | 6/6,155 | 0.7096 | 0.2459 | 0.8382/0.6709/0.7520 | 0.9079 | Baseline | 1.13 s / 5,676 MB | 2.23% |
| | **GSPA Stage 1+2A** | 6/6,155 | 0.7117 | 0.2453 | 0.8380/**0.6732**/**0.7555** | 0.9079 | +0.0021 [−0.0006, +0.0085] | 6.67 s / 4,232 MB | 2.23% |
| Viruses | Stage 0 Noisy-OR | 20/2,805 | 0.7222 | 0.3112 | 0.8303/0.7726/0.7004 | 0.7576 | Baseline | 0.27 s / 4,841 MB | 1.90% |
| | **GSPA Stage 1+2A** | 20/2,805 | 0.7252 | 0.3116 | **0.8304**/**0.7747**/**0.7037** | 0.7576 | +0.0030 [+0.0004, +0.0067] | 0.42 s / 4,575 MB | 1.90% |

# S4 Secondary metabolism, metagenomic dark matter, and Stage 2B stoichiometric refinement

In the Price-149 enzyme evaluation, the cognate-substrate stoichiometric neighborhood probe and outer-loop damped anchor are defined as follows. For each candidate pair $(p, u)$, let $n_1(p, u)$ and $n_2(p, u)$ count chromosomal neighbors within $\pm 10$ genes whose reactions share non-currency metabolites with $u$ at 1 and 2 hops, $f_{\mathrm{op}}(p)$ denote the co-directional operon fraction, and $s_{\mathrm{nb}}(p, u) = \min(2.0, \sum_{|g'-g|\le 10}(1 + 0.25|g' - g|)^{-1} P_{\mathrm{cand}}(g', u))$ measure distance-weighted neighbor EC support. The local stoichiometric probe shift $\Delta L_{\mathrm{probe}}(p, u) = 0.85 \min(1.5, 0.55n_1 + 0.20n_2) + 0.45 f_{\mathrm{op}}(p)\mathbb{I}[n_1 + n_2 > 0] + 0.35\mathbb{I}[u \in \mathcal{R}_{\mathrm{bal}}] - 0.50\mathbb{I}[u \notin \mathcal{R}_{\mathrm{ec}}]$ and Stage 1 neighbor coherence shift $\Delta L_{\mathrm{stage1}}(p, u) = 0.90 s_{\mathrm{nb}}(p, u) + 0.50 f_{\mathrm{op}}(p)\mathbb{I}[s_{\mathrm{nb}}(p, u) > 0.20] + 0.30\mathbb{I}[u \in \mathcal{W}_{\mathrm{kegg}}]$ are applied over $T = 3$ damped outer iterations ($\lambda_t \in \{0.70, 0.42, 0.252\}$):

$$P^{(t)}_{\mathrm{single}}(p, u) = \sigma_{\log}\Big(0.75 \operatorname{logit}\big(P^{(t-1)}(p, u)\big) + 0.25 \operatorname{logit}\big(P^{(0)}(p, u)\big) + \lambda_t\big(\Delta L_{\mathrm{stage1}}(p, u) + 0.65\, \Delta L_{\mathrm{probe}}(p, u)\big)\Big), \quad \text{(S1)}$$

and the multi-predictor posterior blends $0.55 P^{(t)}_{\mathrm{single}}(p, u) + 0.45 P_{\mathrm{NoisyOR}}(p, u)$. Supplementary Table S10 reports the biosynthetic gene cluster benchmark across 106 MIBiG/antiSMASH clusters, Supplementary Table S11 reports the 97-MAG uncultured microbial dark-matter panel across 16 phyla, and Supplementary Table S12 reports the 108-genome stoichiometric invariant audit, the comparison across six manually curated BiGG reference genome-scale metabolic models, and the role-stratified Price-149 breakdown on CLEAN.

**Table S10. Biosynthetic gene cluster (BGC) recovery and non-cluster chromosomal specificity across 106 curated clusters from 12 bacterial species.**

| BGC Class | Clusters ($n$) | Core / Tailoring Loci | Stage 0 Core Rec. | GSPA Core Rec. | Stage 0 Tailoring Rec. | GSPA Tailoring Rec. | Non-BGC Specificity / Overall $F_1$ (Stage 0 → GSPA) |
|---|---|---|---|---|---|---|---|
| NRPS (Non-ribosomal peptide synthetase) | 38 | 114/266 | 0.7807 | 0.9211 | 0.5414 | 0.8383 | 99.58% / 0.7420 → **0.8962** |
| Type I / II PKS (Polyketide synthase) | 29 | 87/203 | 0.7586 | 0.9080 | 0.5222 | 0.8227 | 99.62% / 0.7227 → **0.8865** |
| Terpene / Carotenoid | 18 | 36/90 | 0.8056 | 0.9444 | 0.5667 | 0.8556 | 99.69% / 0.7619 → **0.9136** |
| Siderophore (Aerobactin-like) | 12 | 36/72 | 0.8333 | 0.9722 | 0.5972 | 0.8889 | 99.72% / 0.7935 → **0.9384** |
| RiPP / Bacteriocin / Lantipeptide | 9 | 18/54 | 0.7222 | 0.8889 | 0.4815 | 0.7963 | 99.55% / 0.6842 → **0.8613** |
| **All 106 BGCs** | 106 | 291/685 | 0.7801 | 0.9244 | 0.5387 | 0.8380 | 99.62% / 0.7395 → 0.8959 |

**Table S11. Pathway gap resolution across 97 high-quality metagenome-assembled genomes (MAGs) of uncultured bacteria and archaea spanning 16 phyla.**

| Phylum / Group | Domain | MAGs ($n$) | Candidate Fills ($q \geq 0.30$) | High-Conf. Fills ($q \geq 0.50$) | Mean Fills / MAG ($q \geq 0.30/0.50$) | Pathway Comp. Before | Pathway Comp. After |
|---|---|---|---|---|---|---|---|
| Pseudomonadota | Bacteria | 19 | 502 | 137 | 26.4/7.2 | 0.542 | 0.684 |
| Actinomycetota | Bacteria | 12 | 298 | 82 | 24.8/6.8 | 0.518 | 0.661 |
| Bacteroidota | Bacteria | 11 | 237 | 61 | 21.5/5.5 | 0.491 | 0.642 |
| Bacillota | Bacteria | 9 | 208 | 58 | 23.1/6.4 | 0.528 | 0.669 |
| Cyanobacteriota | Bacteria | 8 | 158 | 39 | 19.8/4.9 | 0.534 | 0.675 |
| Chloroflexota | Bacteria | 7 | 143 | 36 | 20.4/5.1 | 0.462 | 0.618 |
| Halobacteriota | Archaea | 7 | 132 | 32 | 18.9/4.6 | 0.489 | 0.631 |
| Thermoproteota | Archaea | 6 | 105 | 25 | 17.5/4.2 | 0.495 | 0.638 |
| Other 8 Phyla (CPR, Asgard, etc.) | Mixed | 18 | 392 | 98 | 21.8/5.4 | 0.469 | 0.613 |
| **All 97 Uncultured MAGs** | **All (16 Phyla)** | 97 | 2,175 | 568 | 22.4/5.9 | 0.511 | 0.656 |

**Table S12. 108-genome stoichiometric invariant audit, curated BiGG reference GEM comparison ($n = 6$ species), and role-stratified Price-149 CLEAN refinement across Strata A, B, and C.**

| Panel / Stratum | Method / Reconstruction Pipeline | Active / Growing | Added Rxns (Cert. / Pad.) | Imbal. / Dead-End Added | EC Prec. / Rec. / $F_1$ | BiGG Rxn $F_1$ | Top-1 Hits (Acc.) |
|---|---|---|---|---|---|---|---|
| 108-Genome Audit | Raw Sequence Threshold ($\geq 0.50$) | 2,809.5 (0% grow) | 0.0 (0.0/0.0) | 2,621.7 active / 9.75 | n/a | n/a | n/a |
| | METEOR v8 (Negative-Cost MILP) | 3,121.0 (100% grow) | 313.35 (51.01/262.34) | 2,597.8 active / 2.36 | n/a | n/a | n/a |
| | Grow-Support Threshold Repair (Pos.-Cost LP) | 2,584.8 (100% grow) | 81.31 (81.31/0.00) | 0.0 added / 2.23 | n/a | n/a | n/a |
| | **GSPA Stage 2B (Lexicographic pFBA)** | 2,815.2 (300.6 **flux**, 100% **grow**) | 5.70 (5.70/0.00) | 0.0 **added** / 0.00 | n/a | n/a | n/a |
| 6 Curated BiGG GEMs | CarveMe v1.5.2 (Native BiGG Database) | 6/6 (100%) | n/a | n/a | 0.8217/0.8927/0.8536 | 0.6955 | n/a |
| | Reconstructor (ModelSEED Namespace) | 4/6 (67%) | n/a | n/a | 0.4387/0.8787/0.5844 | n/a | n/a |
| | DeepProZyme + METEOR v8 (Neg.-Cost MILP) | 6/6 (100%) | n/a | n/a | 0.4383/0.8842/0.5851 | 0.2091 | n/a |
| Price-149 Stratum A ($n = 91$) | CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred** | Core Biomass | n/a | n/a | $F_{\max}$: 0.6336 → 0.6405 → 0.6274 → **0.7019** | Macro: 0.4740 → **0.5288** | 46 → 46 → 47 → **52** (57.1%) |
| Price-149 Stratum B ($n = 55$) | CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred** | Catabolic | n/a | n/a | $F_{\max}$: 0.3580 → 0.3753 → 0.3461 → **0.4546** | Macro: 0.3103 → **0.3436** | 13 → 13 → **15** (27.3%) → 13 (23.6%) |
| Price-149 Stratum C ($n = 2$) | CLEAN Base → + Stage 2B Probe → + Single-Pred → + **Multi-Pred** | Secondary | n/a | n/a | $F_{\max}$: 0.5099 → 0.5099 → 0.5175 → **0.5833** | Macro: 0.5000 → 0.5000 | 1 → 1 → 1 → 1 (50.0%) |

# S5 Experimental phenotype prediction in BacDive and statistical testing

In the BacDive benchmark, the five-fold genus-blocked cross-validation assigns each taxonomic genus deterministically to one fold via MD5 hashing. Within each training fold, tool-specific conditional probabilities $p_{g,\mathrm{cal}}, p_{m,\mathrm{cal}}$, assay prevalence $a_{\mathrm{prev}}(y)$, and fold-local pooled probability $p_{\mathrm{pool}}(s, y)$ ($0.38p_{g,\mathrm{cal}} + 0.18p_{m,\mathrm{cal}} + 0.22a_{\mathrm{prev}} + 0.22P_{\mathrm{Track1}}$ for carbon; $0.36p_{g,\mathrm{cal}} + 0.16p_{m,\mathrm{cal}} + 0.22a_{\mathrm{prev}} + 0.26P_{\mathrm{Track1}}$ for enzymes) are estimated strictly from the training split. Supplementary Table S13 reports the per-fold five-fold genus-blocked cross-validation metrics across both panels.

**Table S13. Summary of 5-fold genus-blocked cross-validation in BacDive across carbon-source growth ($n = 2{,}332$ assays, 529 strains) and diagnostic enzymes ($n = 17{,}496$ assays, 3,017 type strains).**

| Assay Panel | Classifier | Fold 0 $F_1$ | Fold 1 $F_1$ | Fold 2 $F_1$ | Fold 3 $F_1$ | Fold 4 $F_1$ | 5-Fold Mean ± s.d. | Pooled $F_1$ (BAcc / ROC-AUC) |
|---|---|---|---|---|---|---|---|---|
| Carbon Growth ($n = 2{,}332$) | BacDive-AI RF Baseline | 0.8709 | 0.8720 | 0.8397 | 0.7174 | 0.8646 | $0.8329 \pm 0.0659$ | 0.8448 (0.8842/0.9462) |
| | DiASPora GBDT | 0.8582 | 0.8819 | 0.8582 | 0.7817 | 0.8238 | $0.8408 \pm 0.0390$ | 0.8483 (0.8879/0.9519) |
| | Traitar L1-SVM | 0.8480 | 0.8765 | 0.8727 | 0.7708 | 0.8653 | $0.8467 \pm 0.0438$ | 0.8571 (0.8888/0.9430) |
| | **GSPA + DiASPora Hybrid** | 0.8830 | 0.9048 | 0.8824 | 0.7778 | 0.8602 | $0.8616 \pm 0.0495$ | 0.8710 (0.9035/0.9567) |
| Diagnostic Enzymes ($n = 17{,}496$) | BacDive-AI RF Baseline | 0.8477 | 0.8371 | 0.8490 | 0.8553 | 0.8387 | $0.8456 \pm 0.0076$ | 0.8458 (0.8427/0.8962) |
| | DiASPora GBDT | 0.8465 | 0.8579 | 0.8560 | 0.8655 | 0.8493 | $0.8550 \pm 0.0075$ | 0.8550 (0.8535/0.9061) |
| | Traitar L1-SVM | 0.8620 | 0.8561 | 0.8562 | 0.8714 | 0.8517 | $0.8595 \pm 0.0076$ | 0.8593 (0.8552/0.9034) |
| | **GSPA + DiASPora Hybrid** | 0.8671 | 0.8663 | 0.8718 | 0.8854 | 0.8653 | $0.8712 \pm 0.0083$ | 0.8713 (0.8714/0.9097) |

# S6 Stage 3 neuro-symbolic tool schemas, multi-seed stability, prospective CAFA6, and extremophile case study

Stage 3 provides five deterministic, read-only genomic and ontological inspection tools to the open-weight instruction model (`run_gspa_agent.py`):

1. `inspect_operon_neighborhood(protein_id, window=5)`: Returns the de-identified genomic neighborhood (±5 genes on the same contig), intergenic distances, strand orientation, and neighbor GO/EC annotations.

2. `inspect_pathway_or_complex_gap(anomaly_id)`: Returns the target pathway or heteromeric complex identifier, fraction of partner steps or subunits present in the genome at $P \geq 0.50$, and list of missing steps or

subunits.

3. `query_protein_evidence(protein_id)`: Returns Stage 0 per-channel calibrated probabilities, Pfam/InterPro structural domain annotations, and precomputed RLGC kernel density Bayes factors ($\mathrm{BF}_{\mathrm{operon}}$).

4. `check_symbolic_constraints(go_or_ec_term, taxon_id)`: Queries the deterministic ontology bridge (26,911 direct GO taxon rules and 2,715 GO–Uberon/PO/CL anatomical rules linked to 8,331 taxon constraints across 7,722 anatomical and 1,818 taxonomic hierarchy edges) and returns whether the term is valid in `taxon_id`, along with its nearest valid non-root GO ancestors if forbidden.

5. `propose_resolution(action, target_protein_id, resolved_term, confidence, justification)`: Submits the final structured decision (PROMOTE, RELAX, DEMOTE, or ABSTAIN), which is deterministically verified by the symbolic guardrail before acceptance.

Supplementary Table S14 reports the per-seed and three-seed summary of the full guarded multi-turn Stage 3 configuration across seeds 42, 43, and 44 on the 120-anomaly and 50-unique-protein LRO benchmarks, Supplementary Table S15 reports the prospective CAFA6 evaluation across 769 targets, and Supplementary Table S16 summarizes the five desert extremophile isolates from the Empty Quarter (*Rub' al Khali*).

**Table S14. Three-seed evaluation of Stage 3 guarded multi-turn reasoning across the 120-anomaly and 50-gap LRO benchmarks.**

| Evaluation Run | Random Seed | Precision | Recall | $F_1$ Score | Accuracy | Violations Accepted | LRO-50 Hit@1 (MRR) |
|---|---|---|---|---|---|---|---|
| Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50) | Seed 42 | 0.9787 | 0.8519 | 0.9109 | 0.9250 | 0/120 (0.0%) | 82.00% (0.8740) |
| Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50) | Seed 43 | 0.9592 | 0.8704 | 0.9126 | 0.9333 | 0/120 (0.0%) | 84.00% (0.8890) |
| Stage 3 Multi-Turn + Guardrail (120 Anomalies + LRO-50) | Seed 44 | 0.9787 | 0.8519 | 0.9109 | 0.9250 | 0/120 (0.0%) | 82.00% (0.8747) |
| **Stage 3 Multi-Turn + Guardrail (3-Seed Mean ± s.d.)** | **Seeds 42, 43, 44** | $0.9722 \pm 0.0092$ | $0.8581 \pm 0.0087$ | $0.9115 \pm 0.0008$ | $0.9278 \pm 0.0039$ | $0.00 \pm 0.00\%$ | $82.67 \pm 0.94\%$ ($0.8792 \pm 0.0069$) |

**Table S15. Prospective CAFA6 evaluation across 769 targets (412 no-knowledge and 357 limited-knowledge targets).**

| Method | Targets ($n$) | MF $F_{\max}$ ($\mathrm{WF}_{\max}$) | BP $F_{\max}$ ($\mathrm{WF}_{\max}$) | CC $F_{\max}$ ($\mathrm{WF}_{\max}$) | Macro $F_{\max}$ | Macro $\mathrm{WF}_{\max}$ | NK Macro $F_{\max}$ | LK Macro $F_{\max}$ | Runtime / Peak RAM |
|---|---|---|---|---|---|---|---|---|---|
| ESM-2 + ProstT5 3Di MLP (2025) | 769 | 0.4520 (0.3840) | 0.3700 (0.3420) | 0.6650 (0.5860) | 0.4957 | 0.4370 | 0.4612 | 0.5355 | 18.4 s / 5,200 MB |
| Agentic AFP + Domain/EC (Qwen3-14B, 2026) | 769 | 0.4810 (0.4100) | 0.3990 (0.3700) | 0.6590 (0.5590) | 0.5130 | 0.4463 | 0.4795 | 0.5516 | 24.0 s / 6,800 MB |
| **GSPA Stage 3 Neuro-Symbolic Agent** | 769 | 0.5470 (0.4980) | 0.4140 (0.3720) | 0.6690 (0.5820) | 0.5433 | 0.4840 | 0.5124 | 0.5790 | 19.8 s / 5,400 MB |

**Table S16. Empty Quarter (*Rub' al Khali*) desert extremophile isolate annotation summary across five bacterial genomes.**

| Empty Quarter Isolate | Taxonomic Clade | Proteins ($N$) | Raw $C_{\mathrm{ess}}$ | Stage 0 $C_{\mathrm{ess}}$ | GSPA $C_{\mathrm{ess}}$ | Path. Coh. (Raw → GSPA) | pFBA Repairs / Stage 3 Prom. |
|---|---|---|---|---|---|---|---|
| *Salinicola* sp. EQ-01 | Gammaproteobacteria | 3,842 | 0.8810 | 0.9120 | 0.9680 | 0.4820 → 0.6140 | 6 repairs / 5 promotions |
| *Halomonas* sp. EQ-02 | Gammaproteobacteria | 4,118 | 0.8920 | 0.9210 | 0.9740 | 0.4950 → 0.6280 | 5 repairs / 6 promotions |
| *Pseudomonas* sp. EQ-03 | Gammaproteobacteria | 5,490 | 0.9140 | 0.9380 | 0.9820 | 0.5190 → 0.6410 | 4 repairs / 4 promotions |
| *Bacillus endophyticus* EQ-04 | Bacillota | 4,625 | 0.9010 | 0.9280 | 0.9760 | 0.5040 → 0.6320 | 5 repairs / 7 promotions |
| *Streptomyces* sp. EQ-05 | Actinomycetota | 7,184 | 0.8540 | 0.8960 | 0.9590 | 0.4460 → 0.5980 | 8 repairs / 9 promotions |
| **5-Isolate Mean** | **3 Phyla / Classes** | 5,051.8 | 0.8884 | 0.9190 | 0.9718 | 0.4892 → 0.6226 | 5.6 **repairs** / 6.2 **promotions** |